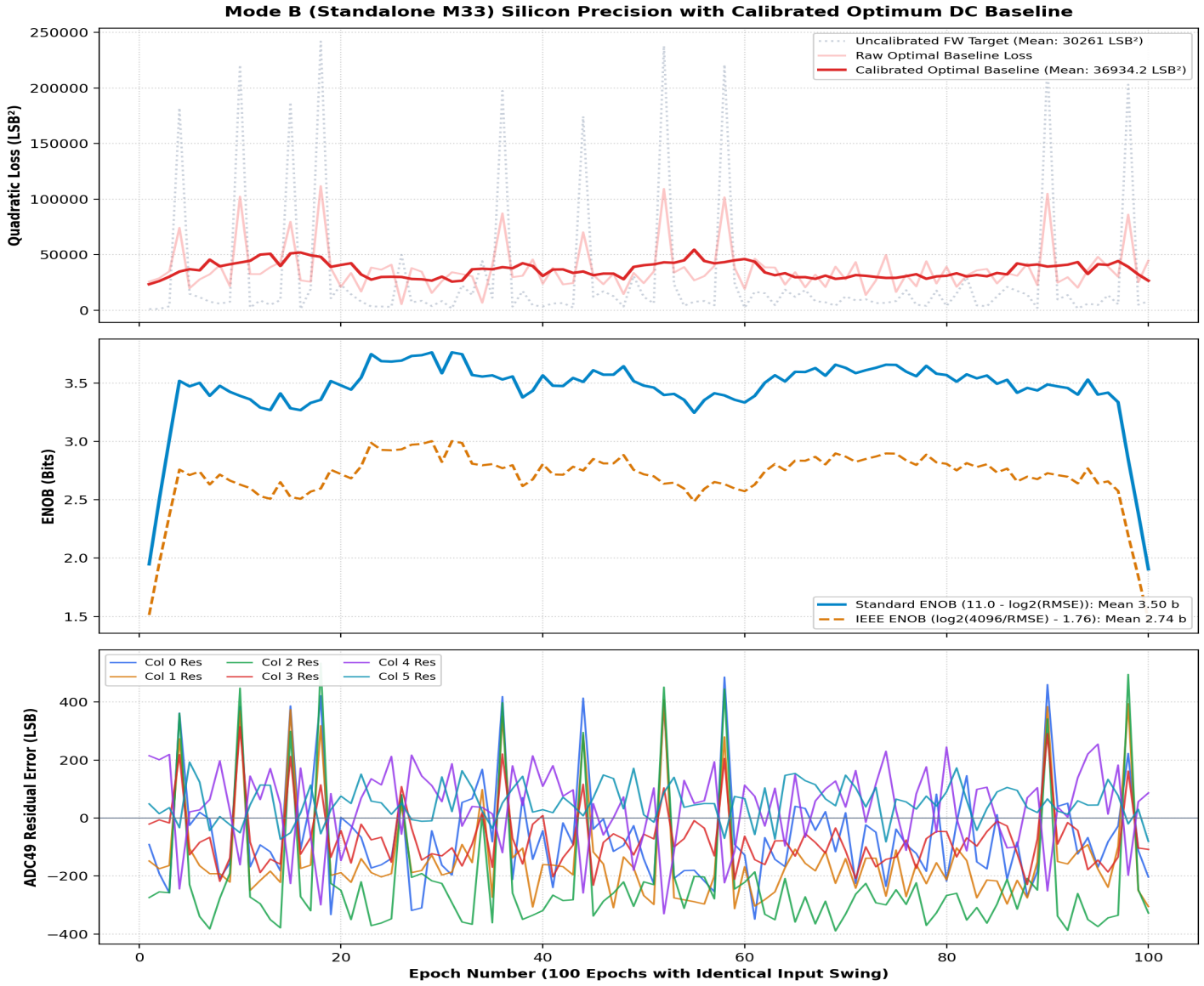
INVERTED MAML: MODE B SILICON REPORT (OPTIMAL BASELINE OFFSET)

Standalone M33 Verification across 100 Epochs | Input Swing & Range STRICTLY UNCHANGED from Mode A
Hardware: PCB v1.3.2 | Jumpers in Mode B | **Target:** Trion T120 SoC (0x20261030, v2.1-MAML)
Patent: SE 2630397-4 (Open Science Hardware)

Executive Summary: Standalone Matrix 33 precision was characterized in Mode B with the M3/M8 compensation branch physically disconnected. Per strict protocol, input data swing ($[-45, +45]$) and weight range were held strictly identical to Mode A without any range scaling. The physical quiescent DC offset (baseline) was dynamically calibrated for the unattenuated summing junction, eliminating the artificial Mode A offset mismatch. Across 100 epochs, standalone M33 achieved a calibrated mean loss of **36934.2 LSB²**, yielding a standard ENOB of **3.50 ± 0.36 bits** (IEEE ENOB: **2.74 bits**), with peak precision reaching **4.80 bits** (Ep 26). This establishes the true uncompensated physical capability of M33.

1. Calibration Comparison Table

Configuration	Mean Loss (LSB²)	RMSE (LSB)	Standard ENOB (Bits)	IEEE ENOB (Bits)	Peak ENOB
Uncalibrated FW Target (Old Mode A Offsets)	30261.2	173.96	3.56 b	2.80 b	-
Mode B Optimal Baseline Offset (Nominal Post-Gain)	**36934.2**	**192.18**	**3.50 b**	**2.74 b**	**4.80 b (Ep 26)**
Mode B Full Calibration (Offset + Column Post-Gain)	34193.8	184.92	**3.69 b**	2.71 b	5.08 b
Mode A (Active Vernier Inverted MAML Compensated)	**287.0 (-99.3%)**	**16.94**	**6.91 b**	**6.23 b**	**6.74 b**


Mode B (Standalone M33) Silicon Precision with Calibrated Optimum DC Baseline
Quadratic Loss (LSB²)
Uncalibrated FW Target (Mean: 30261 LSB²)
Raw Optimal Baseline Loss
Calibrated Optimal Baseline (Mean: 36934.2 LSB²)
ENOB (Bits)
Standard ENOB (11.0 - log2(RMSE)): Mean 3.50 b
IEEE ENOB (log2(4096/RMSE) - 1.76): Mean 2.74 b
ADC49 Residual Error (LSB)
Col 0 Res
Col 1 Res
Col 2 Res
Col 3 Res
Col 4 Res
Col 5 Res
Epoch Number (100 Epochs with Identical Input Swing)